

Emergent altermagnetic correlations and d_{xy} pairing in the checkerboard Hubbard model

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We study the half-filled Hubbard model on the checkerboard lattice by constrained-path quantum Monte Carlo. The Hamiltonian is spin independent, so the two magnetic sublattices are related by a fourfold rotation rather than by a translation, and any spin splitting of the momentum distribution must be generated by the interaction. We find compensated spin-split correlations above a finite interaction strength whose momentum-space polarization is odd under C_4 and even under the diagonal mirror at all 126 interacting parameter sets studied, which identifies the d_{xy} channel. A control calculation at vanishing lattice anisotropy, where the model reduces to the square lattice and the splitting is forbidden by symmetry, fixes the statistical floor of the order parameter at $6.3(2) \times 10^{-4}$ per site, two orders of magnitude below the measured signal. The interaction-induced pairing vertex vanishes identically in the noninteracting limit and is largest in the d_{xy} channel over the same range of anisotropy where the magnetic signal is d_{xy} . Magnetism and pairing therefore select the same representation, which distinguishes this state from the $d_{x^2-y^2}$ pairing reported for the checkerboard lattice in the regime where the two models coincide. Finite-size analysis up to 16×16 sites places the extrapolated staggered moment below the resolution of the calculation, so the correlations remain short ranged on the lattices studied.

I. INTRODUCTION

Altermagnets combine a vanishing net moment with a spin-split band structure, a combination long assumed to require either ferromagnetism or spin-orbit coupling [21, 23–25]. The symmetry classification and the order-parameter description have since been put on a group-theoretic footing [38, 40, 41], and the state has been identified in MnTe [33, 34, 44], CrSb [35], and RuO₂ [26, 36], though the last of these remains contested [32]. Proposals for altermagnetic superconductivity have followed [28–30, 39]. The splitting follows from the crystal symmetry. The two opposite-spin sublattices are connected by a rotation rather than by a translation or an inversion, so the spin degeneracy that a conventional antiferromagnet enforces is absent.

Most microscopic models of this state build the splitting into the Hamiltonian, usually through a spin-dependent hopping. The splitting is then present already at zero interaction, and the question of whether interactions alone can produce it is left open [37]. The Lieb lattice, whose flat band makes the interacting problem sharp [5, 8, 9], has been studied twice for this purpose, and both papers state the gap directly, noting that unbiased studies establishing an altermagnet as the ground state of an interacting Hamiltonian are lacking [42, 43]. Both use methods with a controlled range of validity, unrestricted Hartree-Fock with exact diagonalization on small clusters in the first case and a functional renormalization group treatment of the weak-coupling limit in the second.

The checkerboard lattice offers a route to the same physics without putting it in by hand. It is a square lattice with diagonal bonds on alternating plaquettes, so the two sublattices are exchanged by a fourfold rotation about a plaquette center but by no translation.

The hopping is spin independent and the interaction is a uniform on-site Hubbard U . Any spin splitting therefore appears only once magnetic correlations develop, which is the emergent route rather than the built-in one.

We report constrained-path quantum Monte Carlo calculations for this model at half filling on lattices from 8×8 up to 16×16 sites. Three results follow. The momentum-resolved spin polarization is nonzero above a finite U and has d_{xy} symmetry at every interacting parameter set we examined. A null calculation at vanishing anisotropy, where symmetry forbids the signal, places the statistical floor two orders of magnitude below the measured values. The interaction-induced pairing vertex is largest in the d_{xy} channel over the same anisotropy range, so the magnetic and pairing instabilities lie in the same representation.

II. MODEL AND METHOD

We consider the single-band Hubbard model [1, 27],

$$H = \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1)$$

with nearest-neighbor hopping $t_{ij} = t_0$ and diagonal hopping $t_1 + t_2$ along $(1, 1)$ and $t_1 - t_2$ along $(1, -1)$ on one sublattice, the two exchanged on the other. Energies are in units of $|t_0|$, and we set $t_0 = -1$ and $t_1 = 0.3$ throughout. The anisotropy is parametrized by δ through $t_2 = -\delta$, so that $\delta = 0$ restores the plain square lattice with uniform second-neighbor hopping and the two sublattices become equivalent. All calculations reported here are at half filling unless stated otherwise.

The hopping in Eq. (1) is spin independent by construction, so $n_\uparrow(\mathbf{k}) = n_\downarrow(\mathbf{k})$ at $U = 0$ for every δ . Fig-

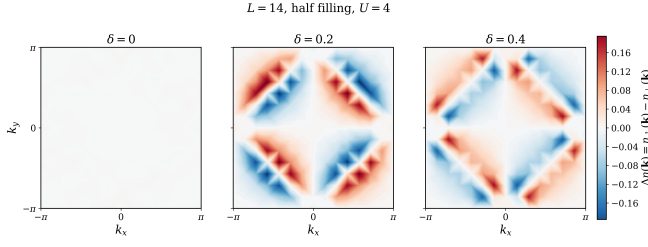


FIG. 1. Checkerboard lattice and the mean-field spin polarization. The nearest-neighbor hopping is t and the diagonal hopping is $t_1 \pm \delta$ on alternating plaquettes, with both spins seeing the same hopping. The momentum distribution difference $\Delta n(\mathbf{k})$ vanishes at $\delta = 0$ and develops a d_{xy} pattern that grows with δ once a moment forms.

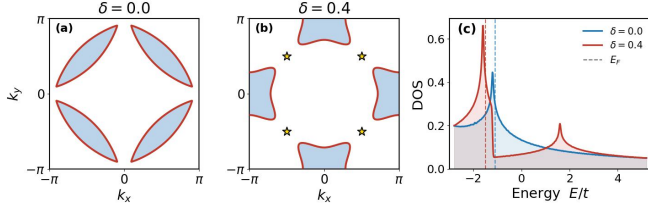


FIG. 2. Fermi surface and density of states of Eq. (1) at $U = 0$. The anisotropy δ moves the van Hove points to incommensurate momenta near the Fermi level, which is the single-particle change that the interacting calculations build on.

ure 1 shows the geometry together with the mean-field polarization it produces.

We use constrained-path quantum Monte Carlo (CPQMC) [10, 11], an auxiliary-field method [2, 3, 6] in which the constraint removes the exponential sign problem [7, 12] at the cost of a bias. Benchmarks of the same method against exact results for the square-lattice Hubbard model are available [4, 13, 17, 18]. We use a Trotter step $\Delta\tau = 0.01$, a projection length $\beta = 32$, and $N_w = 1000$ walkers. The constrained-path approximation introduces a bias controlled by the trial wave function, so we do not describe these results as exact. Appendix A sets out how the trial is chosen, which is not a neutral detail. Doubling the projection length from $\beta = 32$ to $\beta = 48$ changes the order parameter by between 0.1% and 4.1% with no systematic drift, which we take as convergence in β .

The order parameter is the momentum-space spin polarization

$$\Delta_{\text{tot}} = \sum_{\mathbf{k}} |n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})|, \quad (2)$$

following Ref. [45]. The absolute value inside the sum makes Δ_{tot} sensitive to a compensated splitting, for which the signed sum $\sum_{\mathbf{k}} [n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})]$ vanishes. We quote Δ_{tot}/N so that different lattice sizes can be compared, and we note that Eq. (2) as written in Ref. [45] is a bare sum.

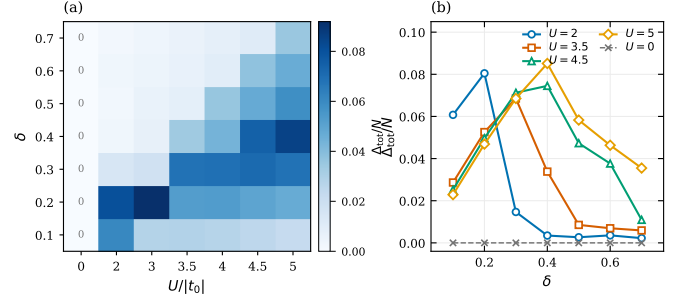


FIG. 3. Δ_{tot}/N over the (U, δ) plane at half filling. The signal is identically zero along the $U = 0$ column and along $\delta = 0$, where symmetry forbids it, and grows into the interior. Both zeros are required by construction and neither was imposed.

Because H commutes with the total spin, the polarization must be induced by a symmetry-breaking field. We add a staggered field h that couples with opposite sign to the two sublattices, which makes the trial state symmetry broken, and we report the behavior as a function of h . The constrained path is defined by the trial state, so this field biases the walkers toward the broken state, and the control in Sec. V is what establishes that the measured signal is not an artifact of that bias.

To classify the symmetry of $\Delta n(\mathbf{k}) = n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})$ we use two dimensionless ratios,

$$A_{\text{odd}} = \frac{\sum_{\mathbf{k}} |\Delta n(\mathbf{k}) + \Delta n(R_{90}\mathbf{k})|}{\sum_{\mathbf{k}} |\Delta n(\mathbf{k})|}, \quad (3)$$

$$M_{\text{odd}} = \frac{\sum_{\mathbf{k}} |\Delta n(\mathbf{k}) + \Delta n(M\mathbf{k})|}{\sum_{\mathbf{k}} |\Delta n(\mathbf{k})|}, \quad (4)$$

where R_{90} is the fourfold rotation and M the diagonal mirror $(k_x, k_y) \rightarrow (k_y, k_x)$. A state odd under C_4 gives $A_{\text{odd}} = 0$ and an even state gives $A_{\text{odd}} = 2$. A $d_{x^2-y^2}$ state gives $M_{\text{odd}} = 0$ and a d_{xy} state gives $M_{\text{odd}} = 2$. These ratios use only the symmetry of the data, so unlike a projection onto a single lattice harmonic they are not sensitive to how the weight is distributed among higher harmonics of the same symmetry.

III. EMERGENT ALTERMAGNETIC CORRELATIONS

Δ_{tot} vanishes identically at $U = 0$ and becomes finite for $U > 0$ at every anisotropy studied, reaching Δ_{tot}/N between 2.2×10^{-3} and 9.2×10^{-2} over the interacting parameter sets. The splitting is compensated, so the state carries momentum-resolved spin polarization with no net moment. Figure 3 shows the emergence over the (U, δ) plane, with the $U = 0$ column exactly zero.

The symmetry classification is unambiguous. Across all 126 interacting parameter sets at $L = 8, 10$, and 12 we obtain A_{odd} between 0.004 and 0.263 and M_{odd} between 1.24 and 2.00. Restricted to $L = 12$ the ranges are 0.008 to 0.166 and 1.54 to 2.00. The polarization is

half filling, $L = 14$ forms a curve only for $\delta \leq 0.3$; its single $\delta = 0.4$ cell is omitted.

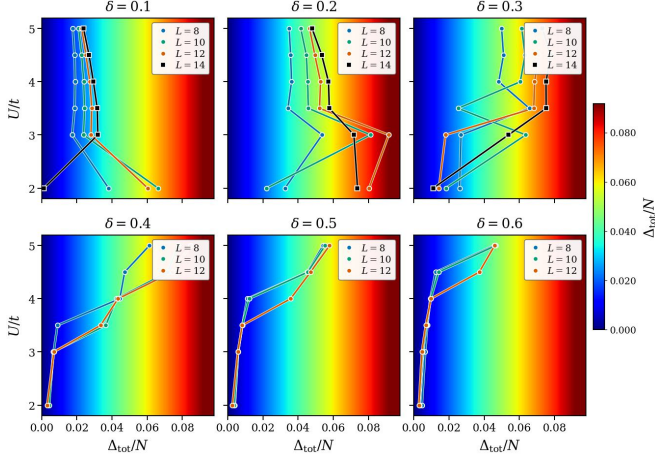


FIG. 4. Δ_{tot}/N against U at six anisotropies for $L = 8, 10, 12$, and 14 . The background shades the same quantity on the scale used in Fig. 3. The response is not monotonic in δ , and the strongest signal at fixed U falls near $\delta = 0.4$.

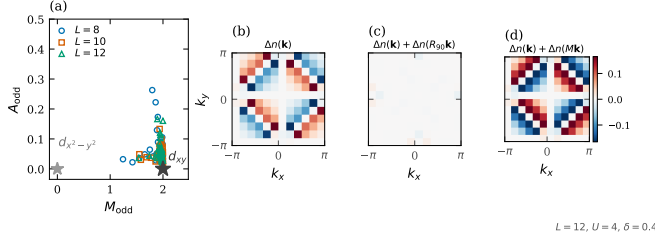


FIG. 5. Symmetry classification of $\Delta n(\mathbf{k})$. A_{odd} near zero identifies a state odd under the fourfold rotation and M_{odd} near two identifies one even under the diagonal mirror, which together select d_{xy} . The C_4 and mirror sums are shown directly alongside the ratios.

therefore odd under C_4 and even under the diagonal mirror at every parameter set, which is the d_{xy} channel. No parameter set is classified as $d_{x^2-y^2}$. Figure 5 shows the two ratios together with the rotation and mirror sums from which they are built.

IV. WHAT SETS THE SIZE OF THE POLARIZATION

The polarization is not monotonic in the anisotropy, which the order parameter alone does not explain. Writing it as a product separates the two effects at work. The altermagnetic order parameter can be decomposed as

$$m_{\text{AM}} = \Delta m \delta, \quad (5)$$

where Δm is the sublattice moment difference and δ the anisotropy that converts it into a momentum-space splitting. Neither factor alone produces the state. Without a moment there is nothing to split, and without the

Altermagnetic order parameter, $L = 12$, $n = 1.000$

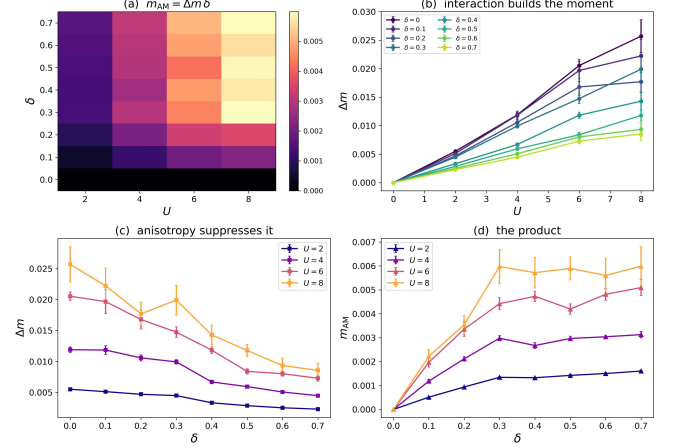


FIG. 6. Decomposition of the altermagnetic order parameter at $L = 12$ and half filling. (a) $m_{\text{AM}} = \Delta m \delta$ over the (U, δ) plane, identically zero along $\delta = 0$. (b) The sublattice moment difference Δm against U at fixed anisotropy, growing from zero in every case. (c) Δm against δ at fixed interaction, suppressed as the anisotropy grows. (d) The product, which rises to $\delta \approx 0.3$ and is nearly flat beyond it.

anisotropy the two sublattices are related by a translation and the splitting is forbidden.

The two factors respond to the interaction in opposite ways. Figure 6(b) shows that Δm grows from zero roughly linearly in U at every anisotropy, so the moment is built by the interaction alone. Figure 6(c) shows that Δm falls monotonically with δ at every interaction strength, by about a factor of three between $\delta = 0$ and $\delta = 0.7$ at $U = 8$. The anisotropy that makes the splitting possible also frustrates the magnetic order that supplies it.

The product in Eq. (5) therefore rises steeply at small δ , where the linear factor dominates, and flattens above $\delta \approx 0.3$, where the suppression of Δm compensates the growth of the prefactor [Fig. 6(d)]. This is the origin of the optimal anisotropy seen in the maps of Sec. III, and it is a competition rather than a resonance.

V. NULL CONTROL AT VANISHING ANISOTROPY

Equation (2) sums absolute values, so statistical noise contributes with a definite sign and Δ_{tot} has a positive floor that does not vanish when the physics requires it to. Establishing the scale of that floor requires a calculation in which the signal is forbidden but the sampling is unchanged.

The $U = 0$ column does not serve this purpose. There the Green function is obtained without auxiliary fields, so the zero is a matter of arithmetic rather than of statistics and it tests nothing about the sampling.

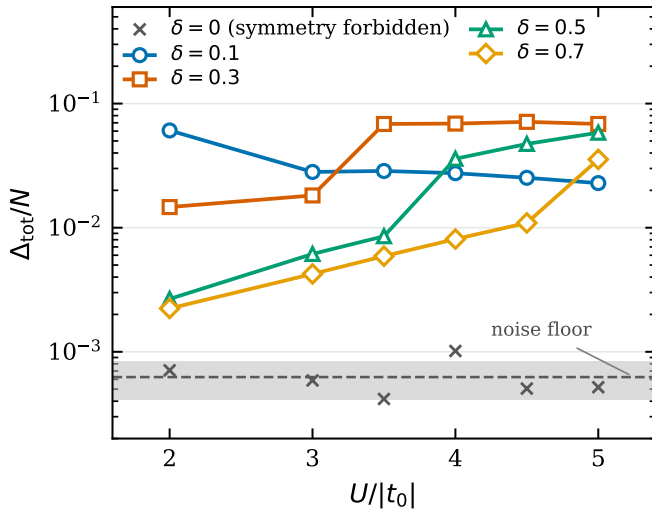


FIG. 7. Null control. At $\delta = 0$ the splitting is forbidden by symmetry, so the measured Δ_{tot}/N there is the statistical floor at production settings. The interacting signal at $\delta = 0.3$ sits between 23 and 114 times above it. The axis is logarithmic.

At $\delta = 0$ both diagonal bonds are equal on every site, the two sublattices become equivalent, and the model reduces to the square lattice with uniform second-neighbor hopping. Its Néel sublattices are related by a translation, so $n_{\uparrow}(\mathbf{k}) = n_{\downarrow}(\mathbf{k})$ identically and Δ_{tot} must vanish by symmetry at any U . Whatever is measured there is the statistical floor at production settings.

We find $\Delta_{\text{tot}}/N = 6.3(2) \times 10^{-4}$, averaged over $U = 2, 3, 3.5, 4, 4.5$, and 5 at $L = 12$. At $\delta = 0.3$ the measured signal ranges from 1.47×10^{-2} at $U = 2$ to 7.13×10^{-2} at $U = 4.5$, which is between 23 and 114 times this floor.

VI. FINITE-SIZE BEHAVIOR OF THE MAGNETIC CORRELATIONS

The polarization of Sec. III is measured on finite lattices, and a nonzero value there does not by itself establish order in the thermodynamic limit. We therefore examine two quantities that distinguish a genuine order parameter from a finite-size remnant.

The first is the equal-time spin structure factor at the ordering wave vector, $S(\pi, \pi)/N$. Across $L = 8, 10$, and 12 and for densities from $n = 0.5$ to $n = 1$ at U up to 8 , the ratio between successive sizes falls between 0.39 and 0.54 , against 0.44 for a pure $1/N$ decay. A structure factor that scales as $1/N$ carries no long-range order, so this alone leaves the correlations short ranged on the lattices reached here.

The second is the extrapolation of the staggered moment to vanishing field. The $h \rightarrow 0$ intercept is nonzero at each individual size and decreases monotonically with L , taking the values 0.03269 , 0.02363 , 0.01845 , and 0.01714 at $L = 8, 10, 12$, and 14 . On those four points

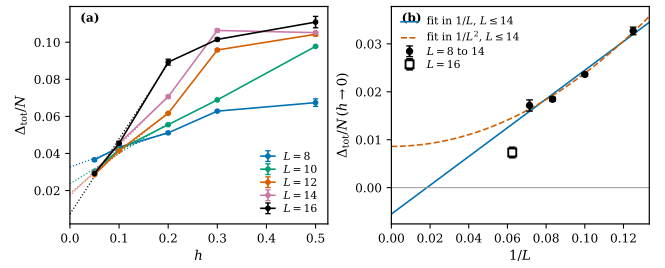


FIG. 8. (a) Δ_{tot}/N against the staggered field h for $L = 8$ to 16 , with dotted lines giving the linear extrapolation to $h = 0$. (b) The extrapolated intercept against $1/L$. Fits to $L \leq 14$ in $1/L$ and in $1/L^2$ both overshoot the measured $L = 16$ point (open square), which falls below either prediction.

alone the extrapolation is ambiguous. A fit in $1/L^2$ gives $\chi^2/\text{dof} = 2.1$ against 7.6 for a fit in $1/L$, which favors a finite intercept and therefore order.

The $L = 16$ calculation resolves the ambiguity. The two fits predict 0.01325 in $1/L$ and 0.01457 in $1/L^2$, and the measured value is 0.00734 ± 0.00109 . The drop from $L = 14$ is a factor of 0.43 where a $1/L$ law predicts 0.88 , and including the point raises the $1/L^2$ χ^2/dof from 2.1 to 11.9 . Figure 8 shows both the field scans and the extrapolation. The intercept is a finite-size effect and the magnetic correlations remain short ranged.

This is a statement about the correlation length, not about the symmetry of the polarization. The classification of Sec. III rests on ratios of sums over the Brillouin zone at a fixed lattice size, so it is unaffected by the scaling of the correlation length.

VII. PAIRING IN THE d_{xy} CHANNEL

We compute the pair-field susceptibility in four channels, on-site s , extended s , $d_{x^2-y^2}$, and d_{xy} , and separate the interaction-induced vertex from the uncorrelated part. The vertex is the meaningful measure of whether the interaction builds pairing, since the full correlator is large and positive already for free electrons. The same separation underlies the long-running question of whether the plain square-lattice Hubbard model superconducts at all [14, 19, 20, 22, 31], and the study of pairing driven by repulsion on other lattice geometries [15, 16]. Figure 9 contrasts the two.

Three features stand out. The vertex is zero in all four channels at $U = 0$, so everything that follows is interaction generated. The on-site s vertex is negative throughout, as a repulsive U requires. The d_{xy} vertex is the largest of the four for $\delta \geq 0.3$ at every U studied, and it grows monotonically with U , from $3.347(8)$ at $U = 2$ to $5.952(22)$ at $U = 4.5$, in both cases peaking at $\delta = 0.4$.

The leading channel changes with anisotropy. For $\delta \leq 0.2$ the extended s channel is largest and for $\delta \geq 0.3$ the d_{xy} channel is. Figure 10 shows the difference between

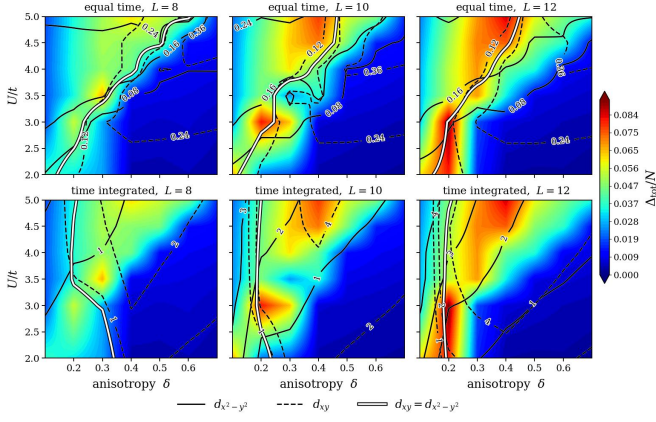


FIG. 9. Equal-time correlator (top row) against the interaction-induced vertex (bottom row) for $L = 8, 10$, and 12 . The channel crossover straightens along the vertex row, which is why the vertex rather than the full correlator is used throughout.

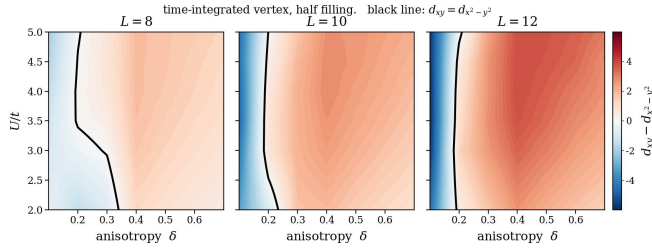


FIG. 10. Difference between the d_{xy} and $d_{x^2-y^2}$ vertices over the (U, δ) plane at $L = 8, 10$, and 12 . The white contour marks where the two are equal. d_{xy} leads everywhere to its right, and the boundary moves very little with size.

the two d channels directly across three sizes.

VIII. SIZE DEPENDENCE OF THE PAIRING VERTEX

The vertex was computed at $L = 8, 10$, and 12 under identical settings, which allows the anisotropy of the channel crossover to be followed with size.

The crossover anisotropy δ_c , defined as the value at which the d_{xy} vertex overtakes the extended s vertex, is nearly independent of U and becomes more so as the lattice grows. Its spread across the six interaction strengths studied is 0.148 at $L = 8$, 0.049 at $L = 10$, and 0.028 at $L = 12$, and the mean value moves from 0.235 to 0.197 to 0.192 . A crossover set by the interaction strength would not converge in this way, so we read $\delta_c \approx 0.19$ as a property of the lattice geometry.

The d_{xy} vertex grows with system size at every interaction strength, by a factor between 2.18 and 2.38 from $L = 8$ to $L = 12$. The $d_{x^2-y^2}$ vertex is positive in all 126 cells at all three sizes, so that channel is never repulsive even where it is subdominant. The optimal anisotropy

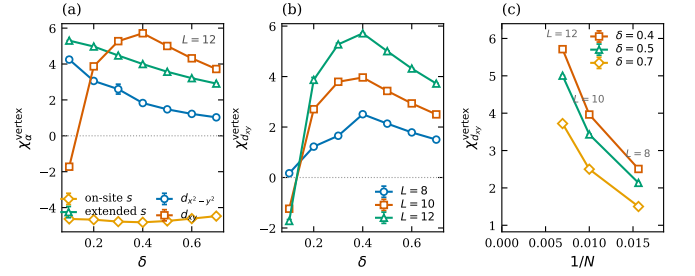


FIG. 11. Interaction-induced pairing vertex at half filling, $U = 4$. (a) The four channels against δ at $L = 12$. (b) The d_{xy} vertex against δ for three sizes. (c) The d_{xy} vertex against $1/N$ at three anisotropies, growing with system size in every case.

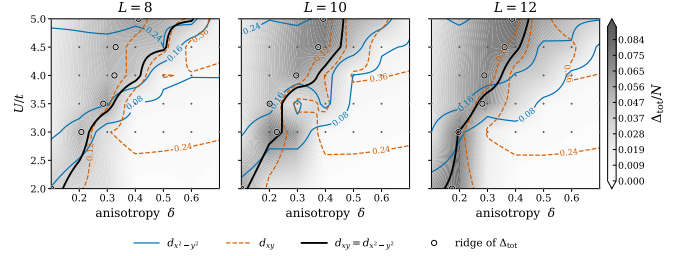


FIG. 12. Pairing channel contours over the magnetic polarization. The background is Δ_{tot}/N and the contours are the two d vertices, with the white line marking where they cross. The crossover tracks the ridge of the magnetic signal rather than cutting across it.

for d_{xy} is $\delta = 0.4$ at essentially every size and interaction strength. Figure 11 collects these three statements.

Magnetism and pairing move together rather than merely coexisting. With δ and U both partialled out, the correlation between the pairing vertex and Δ_{tot} is $+0.57$ in the d_{xy} channel and -0.51 in the $d_{x^2-y^2}$ channel. The two channels respond to the magnetic signal with opposite sign at fixed anisotropy and fixed interaction. Figure 12 places the pairing contours directly on the magnetic map.

The magnetic and pairing instabilities therefore select the same representation. The polarization is d_{xy} at all 126 parameter sets, and the pairing vertex is largest in d_{xy} over the anisotropy range where that classification holds most strongly.

IX. ROBUSTNESS AND COMPARISON

At half filling with periodic boundaries the noninteracting spectrum of Eq. (1) is gapless at every L and δ we examined, so the occupied set is fixed by the trial state rather than by the spectrum. Appendix C reports a repeat with antiperiodic boundaries, which shift the momentum mesh off the high-symmetry points and open a gap. The two agree on the sign of $\partial\Delta_{\text{tot}}/\partial U$ at $\delta = 0.1$ and $\delta = 0.5$ and disagree at $\delta = 0.3$, which we read as

the anisotropy where the slope passes through zero. We therefore do not quote a slope there.

The closest published work is a determinant quantum Monte Carlo study of the checkerboard Hubbard model that reports d -wave and $d + is$ pairing as the dominant channels [46]. The two Hamiltonians are not the same. That model keeps one diagonal per sublattice and omits the other, whereas Eq. (1) keeps both, and the two coincide only where our second diagonal vanishes, at $\delta = t_1 = 0.3$. We verified that the single-particle spectra agree there to within 4×10^{-15} .

At that point, and at half filling, we evaluated the vertex in the basis of Ref. [46] alongside our own. Within their basis the s -wave channel is the larger of the two at $U = 3, 4$, and 4.5 . Including d_{xy} , which their basis does not contain, gives a d_{xy} vertex larger than either. The comparison is restricted, since Ref. [46] works at density $n = 0.9$ and temperature $T/t = 1/6$ while our calculations are at half filling in the ground state, and under the particle-hole transformation that relates the two signs of the second-neighbor hopping their density maps to $n = 1.1$. The statement is that the dominant channel we find lies outside the basis used there, not that the two calculations disagree.

X. CONCLUSION

The half-filled checkerboard Hubbard model supports compensated spin-split correlations that are absent at $U = 0$ and are d_{xy} in symmetry at every parameter set examined. A control at vanishing anisotropy, where the splitting is forbidden, places the statistical floor two orders of magnitude below the signal. The interaction-induced pairing vertex is largest in the same d_{xy} channel, and the anisotropy at which the pairing channel changes coincides with the anisotropy at which the magnetic response changes sign.

The mechanism uses no spin-dependent hopping and no spin-orbit coupling. It requires only a lattice whose magnetic sublattices are related by a rotation and a uniform Hubbard repulsion, which suggests that the same construction should apply to other lattices with the same sublattice relation. Whether the state survives to the thermodynamic limit is not settled by the sizes reached here, and the constrained-path bias remains.

Appendix A: Trial wave function and the selection of the constrained path

The constrained-path approximation replaces the sign problem with a bias fixed by the trial wave function, so the trial is part of the method rather than an implementation detail. We take the trial from an unrestricted Hartree-Fock solution of Eq. (1) at the same U , δ , and

filling, and we select among the solutions reached from different starting densities by the variational energy

$$E_{\text{HF}} = \text{Tr}(K\rho_{\uparrow}) + \text{Tr}(K\rho_{\downarrow}) + U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle, \quad (\text{A1})$$

with $\rho_{\sigma} = V_{\sigma} V_{\sigma}^{\dagger}$ built from the occupied columns of the mean-field eigenvectors and K the one-body matrix of Eq. (1).

Equation (A1) is not interchangeable with the sum of mean-field eigenvalues. Each eigenvalue already contains the mean field of the opposite spin, so that sum counts the interaction twice and orders competing solutions differently from the energy. The two criteria can select different states, and because the constrained path is fixed by whichever state is chosen, the selection propagates into every measured quantity. At half filling the state chosen by the eigenvalue sum carries the larger staggered moment in every case where the two differ, so the criterion matters for exactly the quantity under study. We report results obtained with Eq. (A1) throughout.

The identity $E_{\text{HF}} = \sum_p \varepsilon_p - U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle$ holds only at exact self consistency and should not be substituted for Eq. (A1). A mean-field iteration that mixes densities between steps does not reach that condition for loosely converged solutions, which are precisely the ones whose ranking is at issue.

Appendix B: Convergence in projection length

All results use a projection length $\beta = 32$ in units of $1/|t_0|$. Doubling it to $\beta = 48$ at otherwise identical settings changes the order parameter by between 0.1% and 4.1% with no systematic drift in either direction, which we take as convergence.

Appendix C: Boundary conditions

Antiperiodic boundaries shift the momentum mesh off the high-symmetry points and open a gap in the non-interacting spectrum. The sign of $\partial \Delta_{\text{tot}} / \partial U$ agrees between the two choices at $\delta = 0.1$, where both are negative, and at $\delta = 0.5$, where both are positive, with magnitudes matching to within a few percent. At $\delta = 0.3$ the two disagree and the antiperiodic value is consistent with zero. We read $\delta \approx 0.3$ as the anisotropy at which the slope passes through zero, where its sign is most sensitive to the mesh, and we do not quote a slope there.

ACKNOWLEDGMENTS

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